Aurora Solar Microgrid Pilot

Field Performance Report, Version 1.0

Published March 2025 by the Regional Energy Programme

Executive Summary

The Aurora pilot installed solar and battery microgrids at 3 remote sites. The pilot reduced diesel consumption by 38% compared with the baseline year. Average system uptime reached 97.2%, and total generation was 412,000 kWh over the reporting period.

The estimated payback period is 6.4 years. Total capital expenditure was USD 2,150 thousand.

Key Findings

- Diesel consumption fell by 38%.
- System uptime: 97.2%.
- Payback period: 6.4 years.
- Zephyr-7 inverters performed within specification at all sites.

1. Background and Methodology

Remote communities in the region have historically relied on diesel generators, which are costly to fuel and maintain. The programme tested hybrid solar-battery microgrids to lower fuel use and improve reliability. Each site was instrumented with smart meters that logged generation, load and fuel use every fifteen minutes.

Selection Criteria

- Distance from the main grid greater than 40 km.
- Documented annual diesel use above 20 kilolitres.
- Community agreement and available land for the array.

Table 1: Pilot sites and installed capacity

Site	Terrain	Solar	Battery
Harbor Point	Coastal	120 kW	96 kWh
Ridge Station	Mountain	85 kW	64 kWh
Delta Camp	River delta	60 kW	48 kWh

Data validation followed the programme's two-step review: automated range checks, then manual review by an independent engineer.

2. Results

Monthly diesel use dropped sharply once the arrays were commissioned. The chart below compares kilolitres of diesel consumed per month before and after installation.

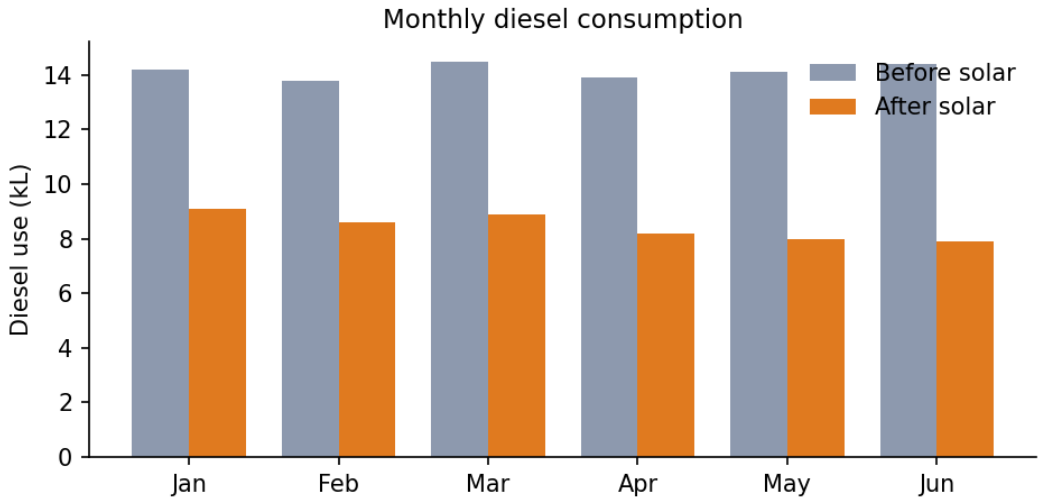

Figure 1: Monthly diesel consumption before and after solar installation, in kilolitres.

Table 2: Energy supplied by source

Month	Total load (kWh)	Solar supplied (kWh)	Solar share
January	58,400	31,200	53%
February	54,900	29,800	54%
March	61,300	34,100	56%

Solar share rose each month as operators tuned battery dispatch schedules.

3. Costs and Risks

Capital expenditure totalled USD 2,150 thousand. The storage units selected were model BATT-4410.

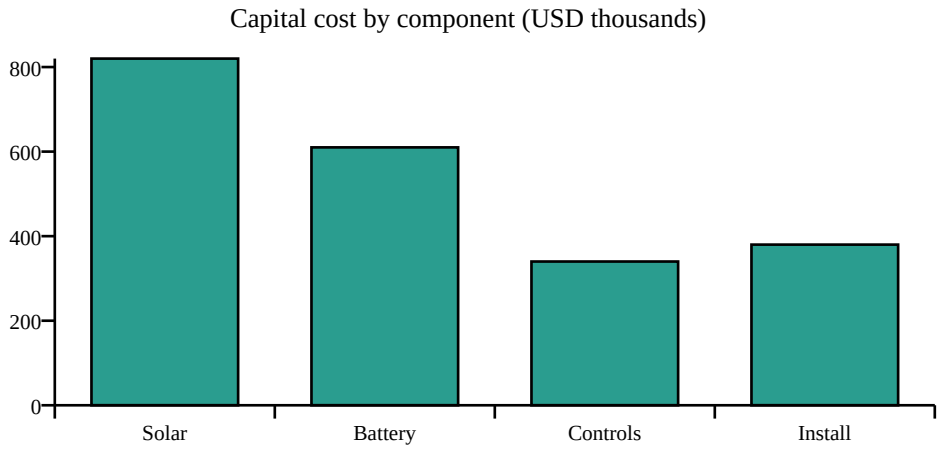

Figure 2: Capital cost breakdown by component, in USD thousands.

Table 3: Capital cost breakdown

Component	Cost (USD k)	Share
Solar array	820	38%
Battery storage	610	28%
Controls and grid	340	16%
Installation	380	18%

Risk Register

- Battery degradation above 3% per year could extend payback.
- Single-vendor dependency for inverters.
- Typhoon season may interrupt maintenance access.

4. Conclusion

Version 1.0 recommends extending the pilot to two additional sites in 2026.

Contact: energy-pilot@example.org. This report contains no information about wind generation or hydrogen storage.